NMK 21103: ELECTROMAGNETIC THEORY

LAB 1
VECTOR ALGEBRA

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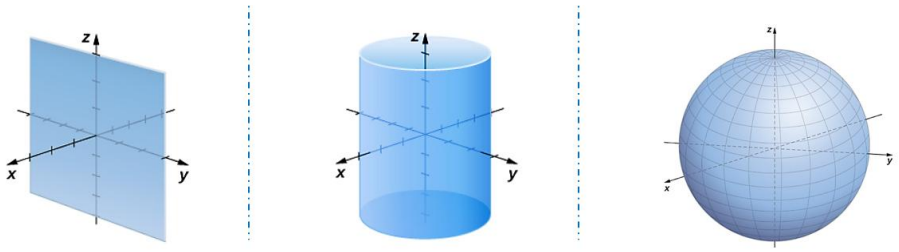
1. Introduction

Vector algebra is a fundamental mathematical tool used in electromagnetic fields analysis. The physical quantities such as electric field intensity, magnetic field intensity, electric flux density, and current density are vector quantities that possess both magnitude and direction. Therefore, the understanding of vector operations and coordinate systems is essential for solving electromagnetic field problems.

The primary coordinate systems consist of **cartesian**, **cylindrical** and **spherical** coordinate as shown in Figure 1.:



(a) Cartesian (x, y, z) (b) Cylindrical (ρ, ϕ, z) (c) Spherical (r, θ, ϕ)

Figure 1

Each coordinate system provides a convenient mathematical representation for analysing electromagnetic fields in structures with specific symmetries. For example, Cartesian coordinates are useful for planar structures, cylindrical coordinates are commonly used in coaxial cables and waveguides, while spherical coordinates are typically used in antenna radiation analysis.

The general vector algebra operations involve **addition**, **subtraction**, **dot product**, and **cross product**. These operations are widely used in Maxwell's equations, field transformations, and electromagnetic energy calculations. This laboratory will introduce the fundamental concepts of vector algebra and coordinate systems using Python.

2. Learning Outcomes

At the end of this laboratory session, students should be able to:

1. Identify and represent vectors in Cartesian coordinate systems.
2. Determine the coordinate representation of vectors in Cartesian, cylindrical, and spherical coordinate systems.

3. Perform vector algebra operations, including vector addition, dot product, and cross product.
4. Perform coordinate transformations between Cartesian, cylindrical, and spherical coordinate systems.
5. Visualize vector quantities in three-dimensional space using Python.

3. Software and Tools

The following tools will be used in this session:

- Python Software
- JupyterLab

Recommended Python packages:

- `vpython`

Run the following code in the first cell

```
try:
    import pkg_resources
except ImportError:
    import pip._vendor.pkg_resources as pkg_resources
    import sys
    sys.modules['pkg_resources'] = pkg_resources

from vpython import *

def add_scene():
    scene = canvas(width=950, height=500, background = vec(0.9,0.9,0.9))

    scene.up          = vec(0, 0, 1)
    scene.camera.pos  = vec(18, 5, 5)
    scene.camera.axis = vec(-20, -4, -4)

    x = arrow(pos=vec(0,0,0), axis=vec(10, 0, 0), shaftwidth = 0.1, color = color.black)
    y = arrow(pos=vec(0,0,0), axis=vec(0, 10, 0), shaftwidth = 0.1, color = color.black)
    z = arrow(pos=vec(0,0,0), axis=vec(0, 0, 10), shaftwidth = 0.1, color = color.black)

    label(pos=x.pos + x.axis, text="x", color=color.black)
    label(pos=y.pos + y.axis, text="y", color=color.black)
    label(pos=z.pos + z.axis, text="z", color=color.black)
```

4. Pre-Laboratory Preparation

Before attending this laboratory session, students should review the following concepts:

1. Definition of vectors and scalars
2. Vector components in Cartesian coordinate systems
3. Vector addition and subtraction
4. Dot product and cross product
5. Coordinate systems used in electromagnetic theory

5. Pre-Lab Questions

Students must answer the following questions before performing the laboratory experiment.

1. Define a vector and provide three examples of vector quantities in electromagnetic theory.

2. Consider $\boldsymbol{A} = \hat{\boldsymbol{a}}_{\boldsymbol{x}} + 2\hat{\boldsymbol{a}}_{\boldsymbol{y}} - 4\hat{\boldsymbol{a}}_{\boldsymbol{z}}$ and $\boldsymbol{B} = -2\hat{\boldsymbol{a}}_{\boldsymbol{x}} + 5\hat{\boldsymbol{a}}_{\boldsymbol{z}}$. Determine:
   (a) $\boldsymbol{A} + \boldsymbol{B}$

   (b) $2\boldsymbol{A} - \boldsymbol{B}$

   (c) $\boldsymbol{A}.2\boldsymbol{B}$

   (d) $\boldsymbol{A} \times \boldsymbol{B}$

Activity 1: Vector Representation in Cartesian Coordinates

A vector $\boldsymbol{A}$ in Cartesian coordinates is expressed as

$$\boldsymbol{A} = A_x\widehat{\boldsymbol{a}}_{\boldsymbol{x}} + A_y\widehat{\boldsymbol{a}}_{\boldsymbol{y}} + A_z\widehat{\boldsymbol{a}}_{\boldsymbol{z}}$$

where

A_x= component along x-axis
A_y = component along y-axis
A_z = component along z-axis

The magnitude of the vector is given by :

$$|\boldsymbol{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Procedure

1. Open JupyterLab.
2. Create a new notebook by clicking on Python 3 (ipykernel)
3. Enter the following code in a cell
4. Run the cell.
5. Observe the graphical output.

```
add_scene()

A = vec(3,-1,-2)

arrow(pos=vec(0,0,0), axis=A, color=color.blue)
label(pos=A, text = 'A', color=color.black)
```

Your Task

i. Calculate the magnitude of vector $\boldsymbol{A}$.
ii. Consider $\boldsymbol{B} = (-5,3,4)$, find the magnitude of vector $\boldsymbol{B}$.

Activity 2: Vector Addition and subtraction

Consider two vectors :

$$\boldsymbol{A} = A_x\hat{\boldsymbol{a}}_{\boldsymbol{x}} + A_y\hat{\boldsymbol{a}}_{\boldsymbol{y}} + A_z\hat{\boldsymbol{a}}_{\boldsymbol{z}}$$

$$\boldsymbol{B} = B_x\hat{\boldsymbol{a}}_{\boldsymbol{x}} + B_y\hat{\boldsymbol{a}}_{\boldsymbol{y}} + B_z\hat{\boldsymbol{a}}_{\boldsymbol{z}}$$

The resultant vector of $\boldsymbol{A} + \boldsymbol{B}$ is

$$\boldsymbol{A} + \boldsymbol{B} = (A_x + B_x)\,\hat{\boldsymbol{a}}_{\boldsymbol{x}} + \left(A_y + +B_y\right)\hat{\boldsymbol{a}}_{\boldsymbol{y}} + (A_z + +B_z)\,\hat{\boldsymbol{a}}_{\boldsymbol{z}}$$

Procedure

1. Define the following vector A and B.
2. Perform vector addition.
3. Plot the vectors.

```
add_scene()

A = vec(3,2,1)
B = vec(1,3,2)

C = A + B

arrow(pos=vec(0,0,0), axis=A, color=color.blue, shaftwidth = 0.3)
label(pos=A, text = 'A', color=color.black)

arrow(pos=vec(0,0,0), axis=B, color=color.green, shaftwidth = 0.3)
label(pos=B, text = 'B', color=color.black)

arrow(pos=vec(0,0,0), axis=C, color=color.red, shaftwidth = 0.3)
label(pos=R, text = 'C', color=color.black)
```

Your Task

Calculate the resultant vector of:

i. $\boldsymbol{A} + \boldsymbol{B}$
ii. $\boldsymbol{B} - \boldsymbol{A}$

Activity 3: Dot Product

The dot product between two vectors is defined as

$$\boldsymbol{A}.\boldsymbol{B} = |A||B|\cos\theta$$

where θ is an angle between vectors.

Note that $$\hat{\boldsymbol{a}}_{\boldsymbol{x}}.\hat{\boldsymbol{a}}_{\boldsymbol{x}} = \hat{\boldsymbol{a}}_{\boldsymbol{y}}.\hat{\boldsymbol{a}}_{\boldsymbol{y}} = \hat{\boldsymbol{a}}_{\boldsymbol{z}}.\hat{\boldsymbol{a}}_{\boldsymbol{z}} = \mathbf{1}$$

while $$\hat{\boldsymbol{a}}_{\boldsymbol{x}}.\hat{\boldsymbol{a}}_{\boldsymbol{y}} = \hat{\boldsymbol{a}}_{\boldsymbol{y}}.\hat{\boldsymbol{a}}_{\boldsymbol{z}} = \hat{\boldsymbol{a}}_{\boldsymbol{z}}.\hat{\boldsymbol{a}}_{\boldsymbol{x}} = \mathbf{0}$$

Procedure

1. Define the following vector
2. Perform the dot product.

```
A = vec(3,4,1)
B = vec(2,1,5)

dot_product = A.dot(B)

magA = A.mag
magB = B.mag

theta = diff_angle(A,B)

theta = acos(dot_product/(magA*magB))
degrees(theta)
```

Your Task

i. Record the results.

ii. Consider $\boldsymbol{C} = (3,0,4)$ and $\boldsymbol{D} = (2,1,2)$, find $\boldsymbol{C.D}$.

Activity 4: Cross Product

The cross product of two vectors produces a vector perpendicular to both vectors.

$$\boldsymbol{A} \times \boldsymbol{B} = \begin{vmatrix} a_x & a_y & a_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \left(A_y.B_z - A_z.B_y\right)a_x - (A_x.B_z - A_z.B_x)a_y + \left(A_y.B_z - A_z.B_y\right)a_z$$

Procedure

1. Define the following vector .
2. Perform the cross-product.
3. Identify the direction of vector **C**.

```
add_scene()

A = vec(1,2,3)
B = vec(3,1,2)

C = A.cross(B)

arrow(pos=vec(0,0,0), axis=A, color=color.blue, shaftwidth = 0.3)
label(pos=A, text = 'A', color=color.black)

arrow(pos=vec(0,0,0), axis=B, color=color.red, shaftwidth = 0.3)
label(pos=B, text = 'B', color=color.black)

arrow(pos=vec(0,0,0), axis=C, color=color.green, shaftwidth = 0.3)
label(pos=C, text = 'C', color=color.black)
```

Activity 5: Cartesian to Cylindrical and Spherical Coordinate Transformation

Conversion from Cartesian (x, y, z) to cylindrical (ρ, ϕ, z) coordinate.

$$\rho = \sqrt{x^2 + y^2} \quad ; \quad \phi = \tan^{-1}\left(\frac{y}{x}\right) \quad ; \quad z = z$$

Conversion from Cartesian (x, y, z) to spherical (r, θ, ϕ) coordinate.

$$r = \sqrt{x^2 + y^2 + z^2} \quad ; \quad \theta = \cos^{-1}\left(\frac{z}{r}\right) \quad ; \quad \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Procedure

1. Consider cartesian coordinate $A = (3,4,5)$.
2. Perform the cylindrical coordinate transformation.
3. Perform the cylindrical coordinate transformation.

```
x = 3
y = 4
z = 5

rho = sqrt(x**2 + y**2)
phi = atan2(y,x)

print(f'rho = {rho}')
print(f'phi = {phi}')
print(f'z   = {z}')
```

```
x = 3
y = 4
z = 5

r     = sqrt(x**2 + y**2 + z**2)
theta = acos(z / r)
phi   = atan2(y, x)

print(f'r = {rho}')
print(f'theta = {theta}')
print(f'phi = {phi}')
```

Your Task

i. Convert the Cartesian coordinates (2,5,2) to cylindrical and spherical coordinate.

ii. If cylindrical coordinate is given by $(5, \frac{\pi}{4}, 3)$, convert into cartesian coordinate.

iii. If spherical coordinate is given by $(6, \frac{\pi}{3}, \frac{\pi}{4})$, convert into cartesian coordinate.

References

Balanis, C. A. (2012). *Advanced engineering electromagnetics* (2nd ed.). John Wiley & Sons.

Sadiku, M. N. O. (2018). *Elements of electromagnetics* (7th ed.). Oxford University Press.

Griffiths, D. J. (2017). *Introduction to electrodynamics* (4th ed.). Cambridge University Press.